Advanced Vector Search & Hybrid RAG Retrieval (Technical Guide)

Abstract:

Retrieval-Augmented Generation (RAG) relies on extracting high-density semantic representations from unstructured documents (PDFs, DOCX files) and ranking them against conversational user queries.

1. Chunking Dynamics and Boundary Preservation:

A naive fixed-character chunking strategy frequently fragments critical numeric tables or splits sentences in half. The optimal approach is recursive character chunking with sliding window overlap (700-1000 characters with 100-150 character overlap). This preserves contextual continuity across semantic boundaries.

2. Dense Embedding vs Sparse Lexical Search:

- Dense Embeddings (Cosine Similarity): Captures synonyms, conceptual equivalence, and cross-lingual meaning.
- Sparse Retrieval (BM25 / TF-IDF): Excels at exact keywords, alphanumeric part numbers, and proper nouns.
- Hybrid Retrieval: Combining dense cosine scores with BM25 keyword matching via Reciprocal Rank Fusion (RRF) achieves superior Recall@5 (94.2%) compared to pure vector search alone (82.1%).

3. Grounding and Verifiable Citations:

To completely eliminate hallucinations, the LLM prompt must mandate source citation anchors [Source N, Page P]. The application UI should cross-link each citation directly to the underlying raw chunk for instant verification.